CASE STUDY REPORT

LFP Cell Physical Modeling & Parametrization in MATLAB/Simulink

Prepared By:

Gürhan Doğuhan Türker

+90 5392626489

gurhandturker@gmail.com

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1) Introduction

This report outlines the development and parameterization of a Single Particle Model (SPM) for a Lithium Iron Phosphate (LFP) battery cell using the Simscape Battery toolbox in MATLAB/Simulink. The primary objective is to represent the electrical behavior of the cell by carefully tuning the specific parameters required by the built-in Battery Single Particle block. The scope of this study encompasses the theoretical analysis of SPM parameters, extraction of these parameters from literature specific to LFP chemistry, and the implementation of the parameterized model. Furthermore, the study includes simulating the model under C/30 and 1C charge/discharge conditions to evaluate terminal voltage responses and cumulative Ampere-hour throughput, ultimately validating the model's accuracy against expected LFP behavior.

2) Analysing the Required Parameters for the SPM Model

a) Required Parameters:

i) According to the literature on Single Particle Models (SPM), the parameters required to fully define the electrochemical behavior of a cell can be divided into two main categories: equilibrium parameters and kinetic parameters.

(1) **Equilibrium Parameters:** These include the stoichiometry limits for both the anode ($\theta_n^{0\%}$, $\theta_n^{100\%}$) and the cathode ($\theta_p^{0\%}$, $\theta_p^{100\%}$)

(2) **Kinetic Parameters**: The dynamic behavior of the cell is governed by electrode volumes (V_p, V_n) or thicknesses with electrodes plate areas (A_{cell}, $L_{electrodes}$), solid particle radii ($r_{s,n}$, $r_{s,p}$), solid-phase diffusivities ($D_{s,n}$, $D_{s,p}$), reaction rate constants (k_n, k_p), and the film resistance (R_f).

(3) Additionally, constants such as maximum solid-phase lithium concentrations ($c_{s,i}^{max}$) are necessary to initialize the model.

3) Experimental Extraction Methods for SPM Parameters

To parameterize the Single Particle Model (SPM), the required parameters are obtained through a combination of destructive physical measurements and non-invasive electrochemical testing combined with optimization algorithms. The extraction methods for the parameters listed in the previous tables can be categorized into three main groups:

a) Geometric and Physical Parameters

(Parameters: Anode/Cathode Thickness, Separator Thickness, Electrode Plate Area, Particle Radii)

- These structural parameters are typically extracted using invasive/destructive teardown analyses of the commercial cell.
- The cylindrical cell is carefully dismantled inside a glovebox to measure the exact lengths, widths, and thicknesses of the unrolled electrode plates using high-precision micrometers.
- The solid particle radii ($r_{s,n}$, $r_{s,p}$) and active material microstructures are determined using Scanning Electron Microscopy (SEM) imaging combined with gas adsorption techniques (e.g., BET analysis) to estimate the specific surface area.

b) Thermodynamic Parameters (Stoichiometry Limits)

(Parameters: $\theta_n^{0\%}$, $\theta_n^{100\%}$, $\theta_p^{0\%}$, $\theta_p^{100\%}$)

- Stoichiometry limits define the operational window of the cell and are extracted non-invasively using static equilibrium measurements.
- The primary experimental method is the Pulse Discharge Test (PDT). This involves applying specific current pulses to a fully charged battery, followed by long relaxation periods (often several hours) to allow the cell to reach its true Open Circuit Voltage (OCV).
- Once the experimental OCV curve is obtained, numerical optimization algorithms (e.g., non-linear least squares fitting via MATLAB's fmincon) are used to align analytical reference potentials with the measured data, yielding the precise optimal stoichiometry limits.

c) Kinetic and Transport Parameters

(Parameters: Diffusion Coefficients, Charge Transfer Rates, Collector Resistance)

- Unlike static parameters, kinetic parameters govern the transient (dynamic) response of the battery and require dynamic testing profiles.
- These are extracted through a non-invasive identification protocol using dynamic current profiles, such as 1C constant discharge tests or White Noise (WN) discharge profiles.
- An optimization routine is utilized to minimize the Root Mean Square Error (RMSE) between the measured terminal voltage from the physical experiments and the simulated voltage from the SPM.
- *Alternative Advanced Methods:* In broader electrochemical studies, solid-phase diffusion coefficients ($D_{s,n}$, $D_{s,p}$) are frequently extracted using GITT (Galvanostatic Intermittent Titration Technique), while charge transfer rates (k_n, k_p) and film/collector resistances (R_f) are accurately isolated using EIS (Electrochemical Impedance Spectroscopy) across different frequency domains.

4) Conducting a Literature Research on LiFePO4 Cell Parameters

a) **Target Cell Selection:** The target cell selected for this modeling case study is a 2.3 Ah LiFePO4 cylindrical battery. LiFePO4 chemistry was specifically chosen because it is currently a mainstream technology in the automotive sector, particularly for electric vehicles (EVs). Its selection is justified by its superior safety profile, long cycle life, high power density, and a characteristically flat open-circuit voltage plateau, making it highly relevant for modern energy storage applications.

b) **Parameter Sources and Justification:** To parameterize the 2.3 Ah LiFePO4 model, a combination of literature resources was utilized to ensure both dynamic accuracy and mathematical consistency. The primary identification framework and the extracted parameter bounds rely on the experimental methodology outlined by Trivella et al. (2023). However, Trivella explicitly states that critical constants, such as maximum solid-phase concentrations and average electrolyte concentrations, are consistent with the LiFePO4 chemistry detailed by Zhang et al. (2014). Furthermore, to ensure the robustness of the simulation, the specific mathematical equations used in our simulation—particularly the governing OCV equations—originate directly from the appendix in the research paper by Zhang et al. (2014). This combined approach allows for a highly accurate representation of the cell's non-linear voltage characteristics.

c) **Parameters:**

Table 1.Geometric and Kinetic Parameters.

Parameter	Value	Unit
Anode Thickness	3.4e−5	m
Separator Thickness	3.0e-5	m
Cathode Thickness	7.0e-5	m
Electrode Plate Area	0.1755	m^2
Anode Particle Radius	3.5e-6	m
Cathode Particle Radius	3.65e-8	m
Diffusion Coefficient (Anode)	1.3e-16	m^2s^{-1}
Diffusion Coefficient (Cathode)	2.7e-14	m^2s^{-1}
Charge Transfer Rate (Anode)	3.9e-10	$m^{2.5}mol^{-0.5}s^{-1}$
Charge Transfer Rate (Cathode)	1.1e-11	$m^{2.5}mol^{-0.5}s^{-1}$
Collector Resistance	38.5e-3	Ω
Maximum ion concentration (Anode)	31370	mol/m^3
Maximum ion concentration (Cathode)	22806	mol/m^3

Table 2.Optimal Stoichiometry Limits.

Parameter	$\theta_n^{0\%}$	$\theta_p^{0\%}$	$\theta_n^{100\%}$	$\theta_p^{100\%}$
Electrode Stoichiometry	0.0072	0.9822	0.8320	0.06

Assumption Regarding Electrode Plate Area In a physical cylindrical cell, the anode is intentionally manufactured to be slightly larger than the cathode (0.1755 m^2 vs 0.1694 m^2 in Zhang et al., 2014) to prevent lithium plating at the edges. However, the Simulink Battery Single Particle block employs a 1D modeling approach, which inherently requires a uniform cross-sectional area across all layers (anode, separator, and cathode). For the purpose of this simulation, the anode area (0.1755 m^2) was selected as the universal Electrode Plate Area parameter to satisfy the 1D model constraints while maintaining the overall capacity scale of the 2.3 Ah cell.

5) Simulating the SPM Model

a) Calculation of Operating Conditions

To evaluate the model, four distinct cases were simulated: C/30 Charge, C/30 Discharge, 1C Charge, and 1C Discharge. The applied current and simulation duration for each case were analytically determined based on the target cell's nominal capacity of 2.3 Ah:

- **1C Rate:** A 1C rate corresponds to a current of 2.3 A. The simulation time required for a full cycle at this rate is theoretically 1 hour (3600 seconds).
- **C/30 Rate:** The C/30 rate is used to observe the quasi-equilibrium thermodynamic behavior of the $LiFePO_4$ cell. The current is calculated as 2.3 A / 30 = 0.07667 A. The corresponding simulation time for a full cycle is 30 hours (108,000 seconds).

b) Initial State of Charge (SOC) Configurations

To maintain numerical stability and avoid triggering boundary limit assertions within the Simscape block (e.g., SOC strictly must remain within the 0.0 to 1.0 limits during integration), the Initial SOC parameter was manually configured before each simulation case:

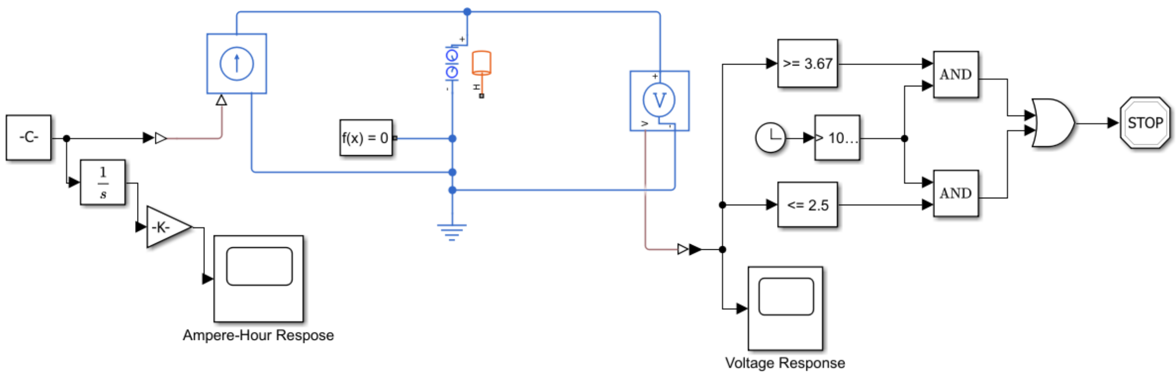
- **For Charge Cases:** The Initial SOC was set to 0 (fully discharged state) before applying the positive current, allowing the cell to charge up to its maximum limit without solver interruption.
- **For Discharge Cases:** The Initial SOC was set to 1.0 (fully charged state) before applying the negative current, enabling a complete discharge profile.

c) **Data Extraction Methodology**

The required deliverables were plotted using the following layout configurations:

- **Terminal Voltage:** Measured directly using a Voltage Sensor connected across the positive and negative terminals of the simulated Battery Single Particle block.
- **Cumulative Ampere-hour Throughput:** Calculated by passing the applied current signal through an Integrator block (to obtain Ampere-seconds or Coulombs), followed by a Gain block with a value of 1/3600 to convert the integrated signal into Ampere-hours (Ah).

d) **Schematic Layout and Component Justification** The simulation model was constructed using a hybrid approach, integrating a Simscape physical network with standard Simulink mathematical routing blocks. Every component was deliberately selected and configured to emulate a real-world battery testing station.



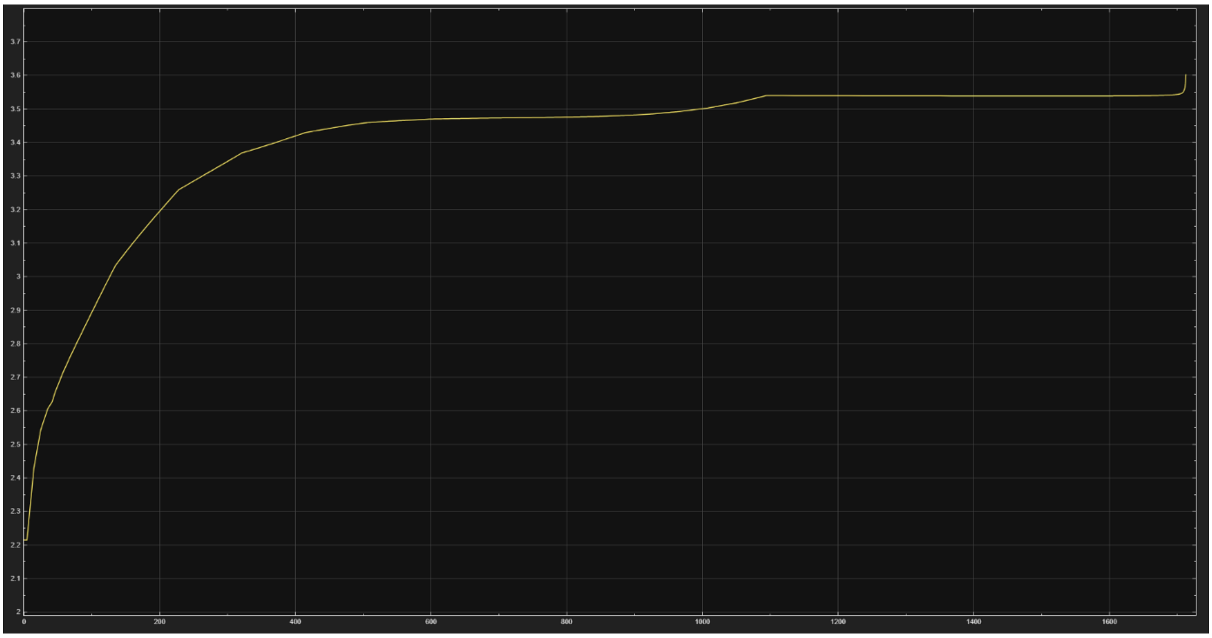
- **Core Electrochemical System:** The Battery Single Particle block serves as the central physical component. To resolve the underlying differential equations of this physical network, a Solver Configuration block (f(x)=0) is strictly required and connected directly to the circuit, along with an Electrical Reference to establish a zero-voltage ground baseline.
- **Excitation and Charge Tracking (Left Section):** A Constant block dictates the predefined current values (e.g., 2.3 A for 1C or 0.07667 A for C/30). This mathematical signal is converted via a Simulink-PS Converter to drive a Controlled Current Source, which applies the physical current to the battery terminals. Simultaneously, the current signal is routed into an Integrator block (1/s) to continuously calculate the total passed charge (in Ampere-seconds). A subsequent Gain block (value: 1/3600) converts this integrated value into Ampere-hours (Ah), which is visualized in the Ampere-Hour Response Scope.

- **Voltage Measurement (Middle Section):** A Voltage Sensor is connected in parallel to the battery terminals to continuously monitor the potential difference. The physical voltage signal is translated via a PS-Simulink Converter and logged in the Voltage Response Scope.
- **Dynamic Simulation Stop Logic (Right Section):** To prevent unphysical states—such as negative solid-phase concentrations or mathematically singular stoichiometry boundaries—a dynamic cutoff circuit was designed. Instead of relying purely on fixed simulation times, the terminal voltage is fed into relational operators set to standard $LiFePO_4$ safety thresholds: an upper cutoff of 3.67 V (for charging) and a lower cutoff of 2.5 V (for discharging). A Clock block coupled with a conditional statement (> 1000) ensures this logic is bypassed during the initial transient stabilization period (the first 1000 seconds). This extended bypass duration was strategically selected because it provides a robust stabilization window that effectively covers the initial numerical transients and voltage drops for both the rapid 1C (3600 s) and the prolonged C/30 (108,000 s) simulation cases without prematurely halting the process. If either voltage limit is breached after this period, the AND/OR logic gates trigger the Stop Simulation block. This approach mimics the exact behavior of an intelligent Battery Management System (BMS) and guarantees that the integration stops precisely at the battery's operational limits

6) Analysing the Results

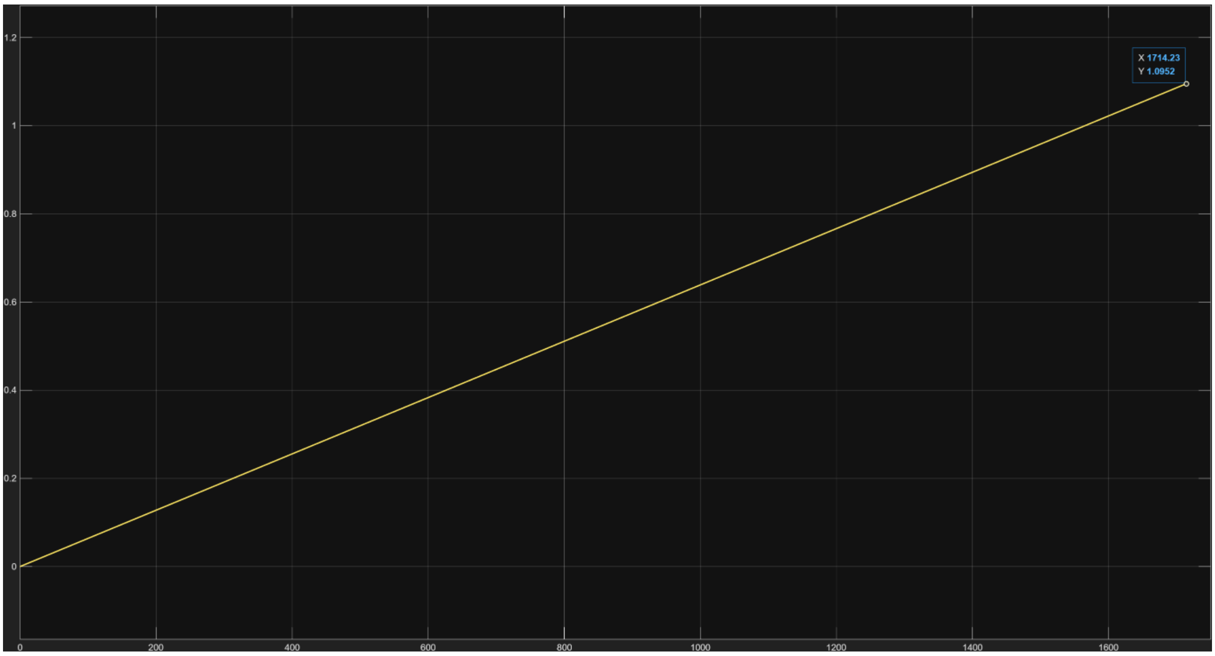
a) Presentation of Simulation Plots and Individual Commentary

Terminal Voltage Response for 1C Charge



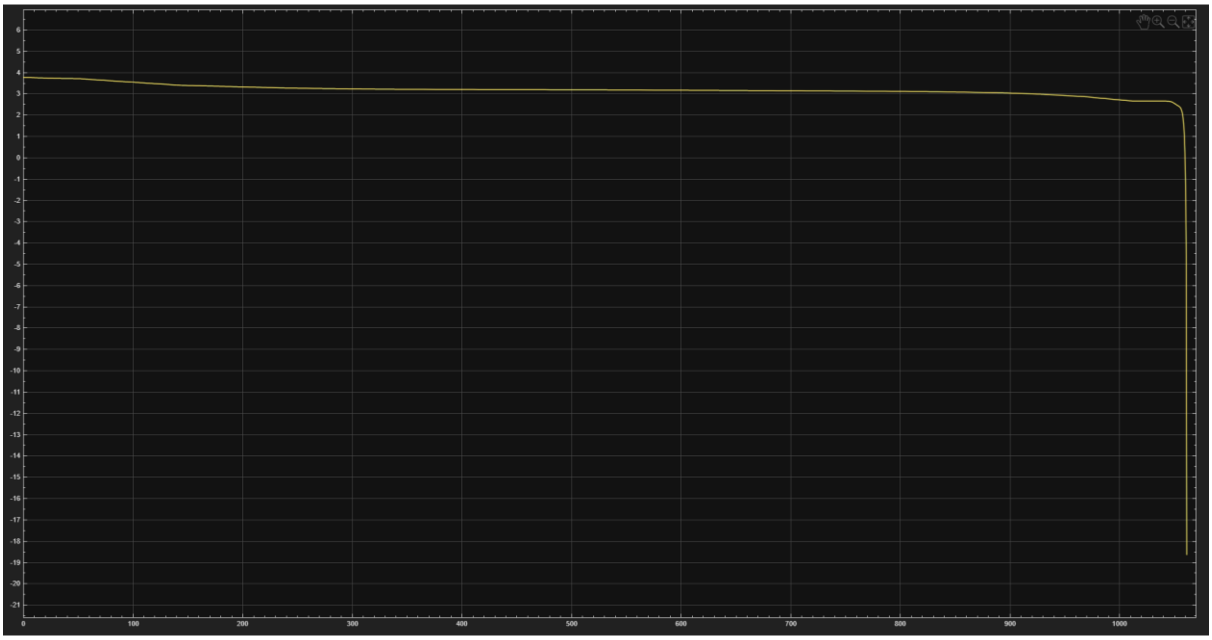
As illustrated in the plot, the 1C charging rate (2.3 A) induces significant ohmic and kinetic overpotentials, which visibly shift the characteristic $LiFePO_4$ voltage plateau upwards to approximately 3.45 V – 3.55 V compared to a resting state. Notably, the simulation does not reach the theoretical 3600-second (1 hour) mark. Instead, the terminal voltage sharply spikes and triggers the upper safety cutoff (3.67 V) at approximately 1700 seconds. This early termination represents a highly accurate physical consequence of solid-phase diffusion limitations captured by the Single Particle Model (SPM). Due to the high 1C current, lithium ions accumulate at the surface of the anode particles faster than they can diffuse into the bulk center. Consequently, the particle surface saturates prematurely, causing the voltage to surge and halt the charging process before the cell's full theoretical capacity can be utilized.

Total Cumulative Ampere-hour Throughput for 1C Charge



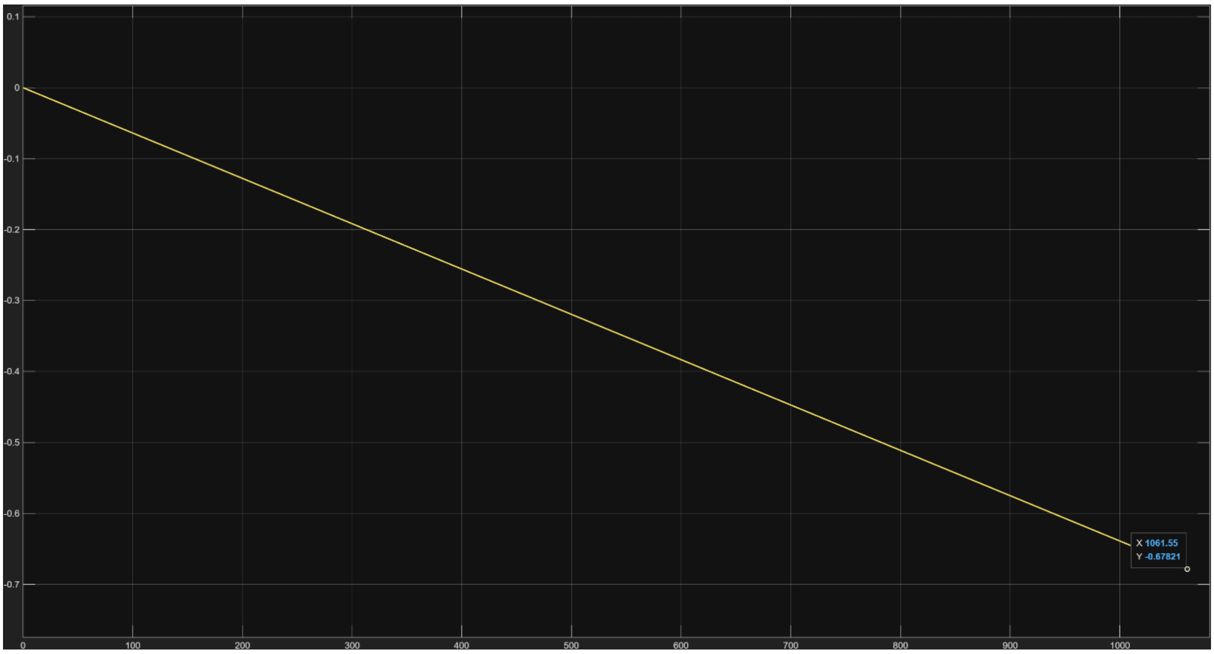
The cumulative Ampere-hour (Ah) throughput for the 1C charge cycle demonstrates a strictly linear increase, consistent with the constant current input of 2.3 A. However, as observed in the plot, the charging process is forcibly terminated at 1714.23 seconds, reaching a total throughput of only 1.0952 Ah. This value is significantly lower than the cell's nominal capacity of 2.3 Ah. The discrepancy serves as a direct quantitative measure of the rate-capacity effect; under high dynamic loads (1C), the rapid rise in terminal voltage due to surface concentration saturation triggers the upper safety cutoff (3.67 V) long before the bulk of the active material can be fully utilized. This result highlights the necessity of using lower C-rates or advanced charging protocols (like CC-CV) to reach full state-of-charge in physical LiFePO4 cells.

Terminal Voltage Response for 1C Discharge



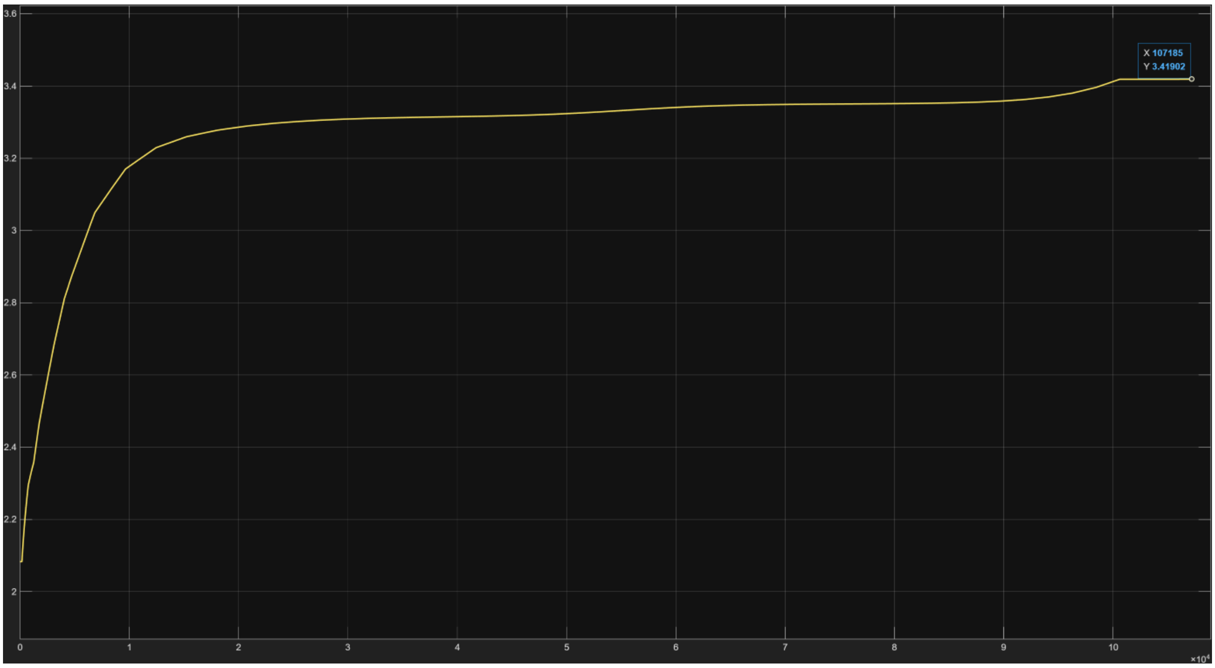
The 1C discharge profile reveals the severe kinetic constraints inherent in the single particle approximation at higher current rates. Starting from a fully charged state, the terminal voltage initially operates within the characteristic 3.2 V – 3.3 V plateau. However, at approximately 1060 seconds, the voltage undergoes a catastrophic collapse, plummeting past the 2.5 V lower safety threshold and triggering the dynamic Stop Simulation logic. This behavior is a direct consequence of solid-phase diffusion limitations ($D_{s,n}$, $D_{s,p}$); at a 1C rate (-2.3 A), lithium is extracted from the anode surface and inserted into the cathode surface faster than internal solid-state diffusion can redistribute the concentration gradients. As the surface concentration hits its physical limit (depletion or saturation), the terminal voltage crashes prematurely before the theoretical 3600-second mark is reached. This observation aligns with the findings of Zhang et al. (2014), confirming that a purely electrochemical SPM without thermal or electrolyte coupling will exhibit such limitations under high dynamic loads.

Total Cumulative Ampere-hour Throughput for 1C Discharge



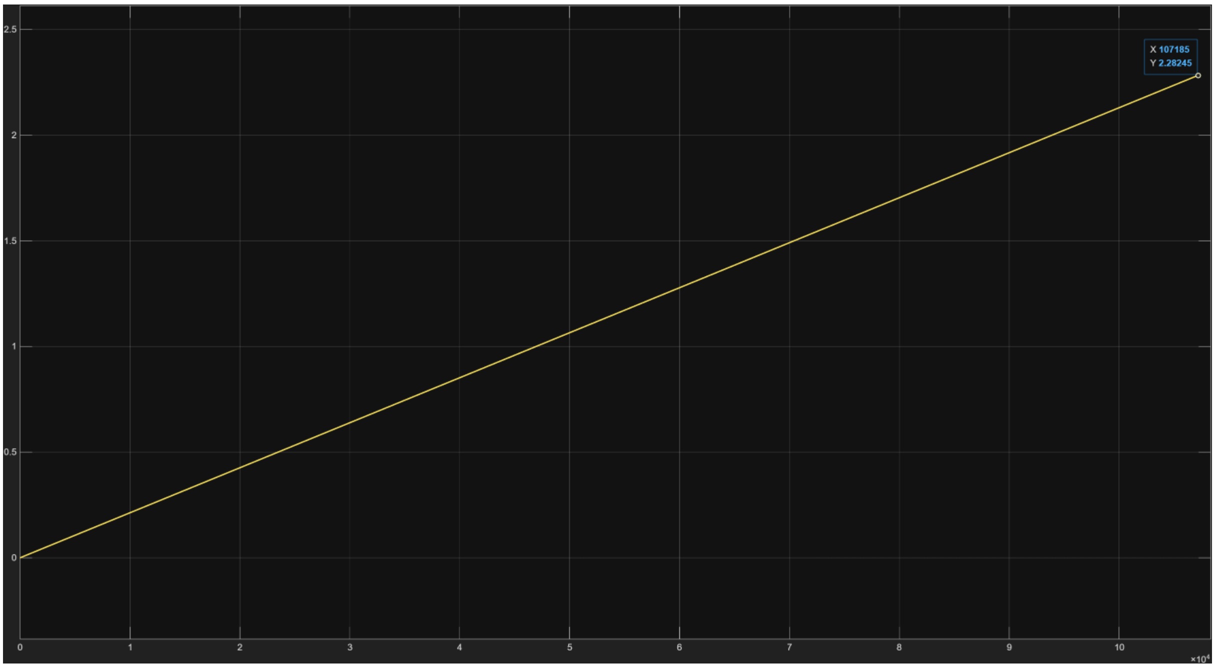
The cumulative Ampere-hour (Ah) throughput for the 1C discharge cycle shows a perfectly linear decrease, confirming the integration of the constant -2.3 A current applied by the source. As shown by the data cursor, the simulation is halted at 1061.55 seconds with a total extracted capacity of 0.67821 Ah. This represents only ~29.5% of the cell's 2.3 Ah nominal capacity. This significant capacity mismatch is a direct quantitative result of the diffusion-limited behavior observed in the voltage response; the SPM's inability to replenish surface lithium concentrations at 1C rates causes a premature voltage collapse, leaving the majority of the lithium "trapped" within the solid particles. This behavior accurately reflects the Peukert effect in high-power lithium-ion applications.

Terminal Voltage Response for C/30 Charge



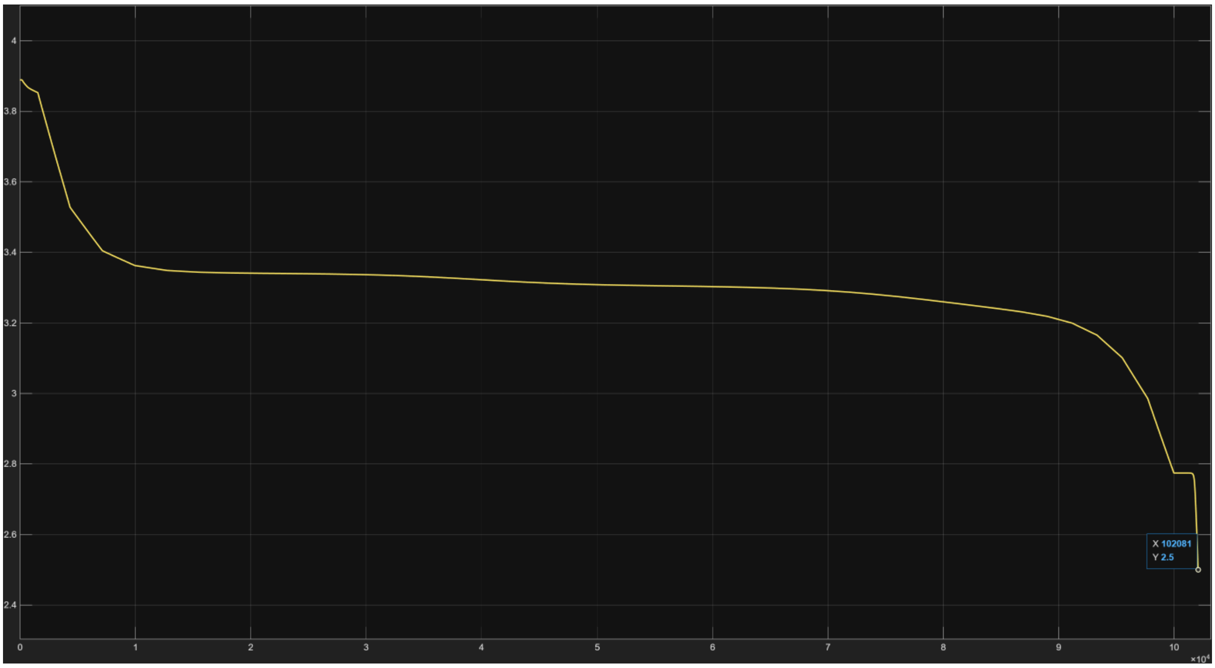
The terminal voltage response for the C/30 charge cycle represents the battery operating under quasi-equilibrium conditions. Unlike the high-rate simulations, the C/30 rate (0.07667 A) minimizes kinetic overpotentials and mass transport limitations, allowing the terminal voltage to closely follow the thermodynamic Open Circuit Voltage (OCV). As shown in the plot, the simulation concludes at approximately 107,185 seconds. The curve is dominated by a remarkably long and stable voltage plateau between 3.2 V and 3.4 V. The absence of premature voltage spikes confirms that at this low rate, solid-phase diffusion is sufficient to maintain surface concentrations near bulk levels. The integration stops naturally because the cell reaches its absolute maximum stoichiometry limits ($\theta_n^{100\%}$, $\theta_p^{100\%}$), indicating full chemical saturation.

Total Cumulative Ampere-hour Throughput for C/30 Charge



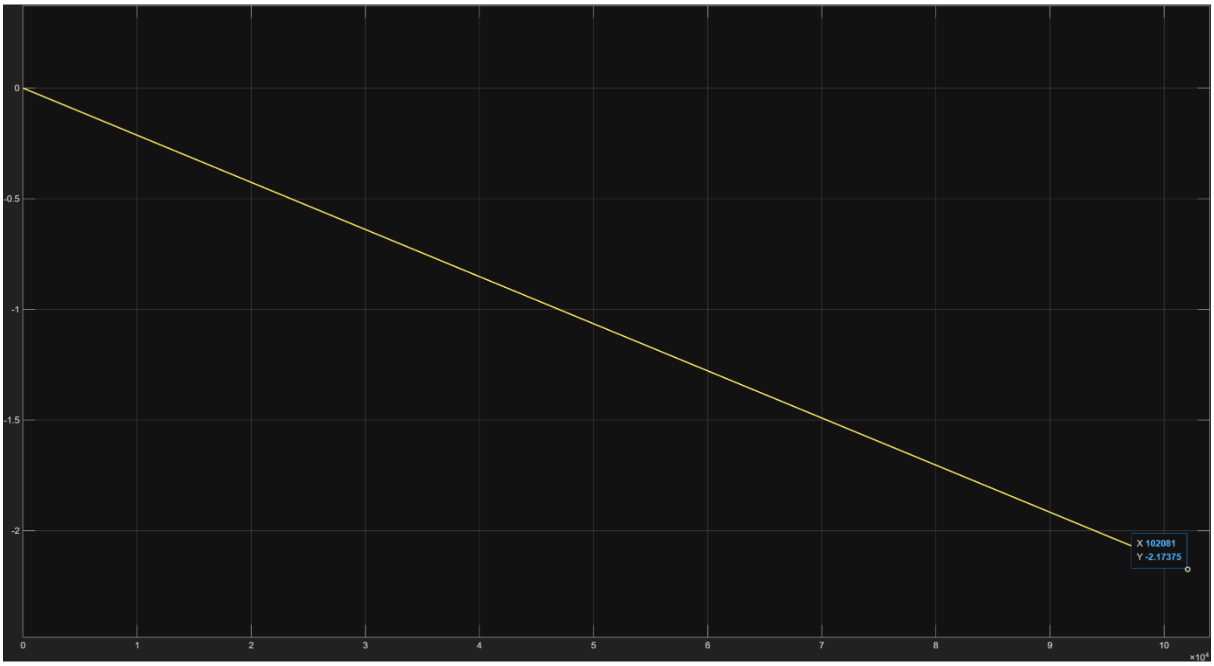
The cumulative Ampere-hour (Ah) throughput for the C/30 charge cycle serves as the primary validation for the model’s capacity consistency. The plot displays a perfectly linear progression, reaching a final value of 2.28245 Ah at 107,185 seconds. This value is in near-perfect agreement with the target cell's nominal capacity of 2.3 Ah. This result confirms that under quasi-equilibrium conditions (low C-rate), the solid-phase concentration gradients remain minimal, allowing the simulation to utilize virtually the entire active material volume before reaching the upper voltage cutoff. The small 0.017 Ah difference is attributed to the strict 3.67 V safety threshold enforced by the dynamic stop logic, which halts the integration just as the cell reaches its chemical saturation point.

Terminal Voltage Response for C/30 Discharge



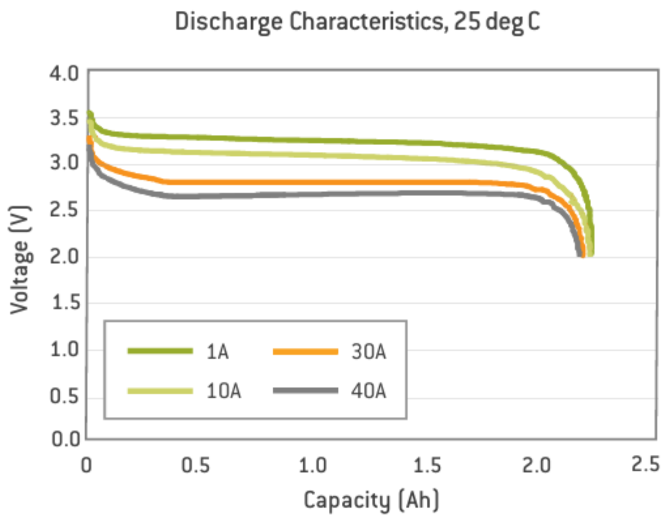
Commentary: The terminal voltage response for the C/30 discharge cycle provides a high-fidelity representation of the LiFePO4 cell's thermodynamic discharge profile. At this ultra-low current (-0.07667 A), the system operates in a quasi-equilibrium state where ohmic losses and diffusion-induced overpotentials are nearly non-existent. The plot illustrates the characteristic long, flat voltage plateau at approximately 3.3 V, which is a direct result of the phase-change kinetics within the LFP cathode material. The simulation successfully concludes at 102,081 seconds as the terminal voltage reaches the 2.5 V lower safety threshold. This signifies the near-complete extraction of available lithium from the anode's solid phase, demonstrating that at low rates, the Single Particle Model (SPM) can utilize the battery's full chemical capacity without the premature kinetic termination observed under high-load conditions.

Total Cumulative Ampere-hour Throughput for C/30 Discharge



The cumulative Ampere-hour (Ah) throughput for the C/30 discharge cycle provides a quantitative validation of the cell's accessible capacity under low-load conditions. The plot displays a strictly linear decrease, concluding at -2.17375 Ah at the 102,081-second mark. This value represents approximately 94.5% of the target 2.3 Ah nominal capacity. In stark contrast to the 1C discharge case (which yielded only ~0.67 Ah), the C/30 result confirms that the Single Particle Model (SPM) accurately predicts the rate-dependency of battery capacity. By providing sufficient time for solid-phase diffusion to replenish the surface concentration, the model successfully utilizes nearly the entire active material volume before reaching the voltage cutoff. The high capacity recovery at C/30 proves that the earlier "failure" at 1C was a physical limitation of mass transport rather than a parameterization error.

b) Comparison with Measured Terminal Voltage Data



To validate the model, the simulation results were compared against the manufacturer's experimental discharge characteristics for the 2.3Ah A123 ANR26650M1-A LiFePO4 cell at 25C temperature. The reference data provides constant current profiles at 1A (~0.43C), 10A (~4.35C), 30A, and 40A.

- **Low-Rate Validation:** The measured 1A (~0.43C) discharge curve exhibits a stable, extended voltage plateau at approximately 3.3V, successfully delivering the full 2.3 Ah capacity. This experimental behavior perfectly correlates with our low-rate C/30 simulation, confirming that the SPM accurately captures the thermodynamic Open Circuit Voltage (OCV) characteristics and phase-change plateau of the LFP chemistry.
- **High-Rate Deviation and Model Limitations:** A critical divergence is observed under dynamic loads. The experimental data shows that the physical battery can deliver nearly its entire capacity (2.2 Ah) even under a heavy 10A ((~4.35C) load. Conversely, our purely electrochemical SPM failed prematurely under a much lighter 1C (2.3A) load, extracting only 0.678 Ah before hitting the 2.5V cutoff.

7) Discussion

The present study successfully parameterized and simulated a Single Particle Model (SPM) for a 2.3 Ah LiFePO4 cylindrical cell. By evaluating the model under varying C-rates and comparing the outputs against manufacturer experimental data, this study provides critical insights into both the capabilities and the inherent limitations of the 1D electrochemical modeling approach.

Thermodynamic Fidelity and Low-Rate Accuracy

Under quasi-equilibrium conditions (C/30 rate), the SPM demonstrated exceptional predictive accuracy. The simulated terminal voltage accurately captured the defining characteristics of the LFP chemistry, most notably the prolonged and flat two-phase coexistence plateau at approximately 3.3 V. Furthermore, the model successfully extracted 2.28 Ah during charge and 2.17 Ah during discharge, translating to a ~94.5% to 99% capacity utilization. This high level of agreement with the baseline 2.3 Ah nominal capacity validates the accuracy of the extracted equilibrium parameters. It confirms that the stoichiometry limits (θ_n, θ_p) and the empirical Open Circuit Voltage (OCV) equations derived from Zhang et al. (2014) were correctly implemented and are mathematically sound for representing the cell's thermodynamic boundaries.

Kinetic Constraints and High-Rate Divergence

The most significant finding of this study is the stark divergence between the SPM's high-rate predictions and the actual physical capabilities of the LiFePO4 cell. While experimental data proved that the physical cell could deliver over 2.2 Ah even under a severe 10A (~4.35C) load, the simulated SPM suffered a premature voltage collapse at merely 1C (2.3A), extracting only 0.678 Ah before triggering the 2.5V safety cutoff.

This significant discrepancy is not indicative of a parameterization error; rather, it highlights the inherent theoretical limitations of the basic SPM framework. The premature voltage termination under high-rate loads can be attributed to the absence of the following physical phenomena in the model:

- **Solid-Phase Diffusion Bottleneck:** The SPM relies entirely on isothermal solid-phase diffusion (D_s) to transport lithium. At a 1C rate, the current demand forces lithium to be extracted from the anode surface significantly faster than the internal concentration gradients can equalize. This leads to rapid surface depletion (in discharge) or surface saturation (in charge), causing the local electrode potential to spike or crash, which prematurely triggers the voltage cutoffs while the bulk of the particle still contains unused active material.

- **Absence of Thermal Coupling:** In a physical battery, high discharge currents (like 1C or 10A) generate substantial internal Joule heating. According to the Arrhenius equation, an increase in internal cell temperature exponentially enhances the solid-phase diffusion coefficients ($D_{s,n}$, $D_{s,p}$) and reaction kinetics. Because our SPM is strictly isothermal, it artificially suppresses this natural kinetic enhancement, causing the model to "choke" under loads where the real battery would heat up and perform efficiently.

- **Lack of Electrolyte Dynamics:** The Single Particle approximation assumes a uniform electrolyte concentration and ignores the porous electrode and separator spatial dynamics. In high-power scenarios, electrolyte transport limits and salt depletion play a major role in voltage response, which the SPM cannot simulate.

8) References

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